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RESEARCH ARTICLE

Toward Intelligent Monitoring in IoT: AI Applications for Real-Time Analysis and Prediction

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ABSTRACT In the contemporary era, the intersection of the Internet of Things and artificial intelligence revolutionizes how industries monitor and optimize their operations. In this work, we present a system that combines real-time monitoring provided by Internet of Things devices with predictive analytics based on artificial intelligence. This system detects anomalies in real-time and anticipates possible failures, allowing proactive interventions to maximize efficiency and minimize operating costs. Our findings reveal a significant improvement in the early detection of abnormal trends, as the system consistently identifies potential problems long before they become critical failures. Our evaluation employed data sets collected from controlled and industrial production environments, with more than 1 million records, including critical parameters such as temperature, humidity, and pressure. The results highlight a significant improvement in the early detection of abnormal trends, with a temperature detection accuracy of 98.7%, exceeding reference values and demonstrating the system's effectiveness in preventing critical failures. The analysis also revealed previously unrecognized operational patterns, offering opportunities for industrial process optimization. This work highlights the effective integration of the Internet of Things and artificial intelligence to improve industrial monitoring, highlighting the tangible benefits of such integration, such as the adaptability and continuous learning of the system, ensuring its long-term effectiveness.

INDEX TERMS Adaptive feedback, real-time precision, intelligent adaptation, IoT-based optimization.

I. INTRODUCTION

In the current era of digitalization, the automation and optimization of processes in various industries have become essential to remain competitive. The adoption of emerging technologies has been a constant in the search to improve efficiency, reduce costs, and guarantee quality [1]. Within this panorama, real-time monitoring of machines and systems has been positioned as a critical tool, especially in sectors where precision and speed in decision-making can make the difference between success and failure. Historically, equipment and processes have been monitored manually, relying heavily on the judgment and experience

of technicians and operators. However, this method has limitations, mainly when dealing with complex or large-scale systems where multiple parameters must be monitored simultaneously [2]. Furthermore, the evolution of Industry 4.0 and the growing demand for intelligent and connected solutions have highlighted the need for more advanced systems. Previous studies have demonstrated the advantages of implementing monitoring tools based on emerging technologies such as the Internet of Things (IoT) and Artificial Intelligence (AI) [3]. These tools enable real-time data collection and offer advanced analysis and prediction capabilities [4].

In response to this scenario, this work proposes to develop an AI-based filtering system specifically designed for real-time event monitoring. Unlike conventional systems,

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this tool continually adapts and learns from the data collected, ensuring superior accuracy and efficiency. It integrates multiple sensors, including IoT cameras and accelerometers, to provide a holistic and detailed view of the operation [5].

Although numerous monitoring systems are on the market, integrating multiple sensors with self-adjustment and learning capabilities represents an innovative proposal. Therefore, this research provides empirical data on the operation and effectiveness of such a system and introduces a novel perspective on how emerging technology can overcome traditional challenges in industrial monitoring and control [6]. This tool detects and predicts trends and patterns, allowing early interventions and avoiding costly downtime or failures. Regarding innovation, the combined use of IoT and AI in our proposal marks a new standard in real-time monitoring, positioning it as one of the appropriate solutions for the current market [7].

After analyzing the existing literature, it is evident that although many researchers have addressed the challenge of real-time monitoring, few have effectively integrated AI adaptability and continuous learning capabilities [8]. Many systems focus on data collection and presentation but lack our tool's deep analysis and prediction capabilities. Furthermore, integrating multiple sensors in our proposal allows a detailed and precise view of the operation, overcoming the limitations of traditional solutions [9].

Unlike previous studies that focus on post-processing performance or IoT systems operating in isolation, our work coherently integrates real-time AI processing with IoT devices. This synergy creates a more robust and efficient system, overcoming the limitations of conventional approaches that did not consider the dynamic interaction between these two components.

This work has demonstrated through practical implementation and constant feedback that the system is theoretically sound, applicable, and valuable in natural environments. Additionally, significant improvements in the efficiency of real-time transmissions have been identified, with notable reductions in interruptions and failures [10]. The system has demonstrated increasing adaptability and accuracy through continuous feedback and learning, validating its effectiveness and potential.

According to the results, this system has immediate practical applications, especially in industries where real-time monitoring is crucial, such as manufacturing, logistics, and critical infrastructure management. By solving problems such as data transmission latency and network overload, our approach not only improves operational efficiency but also reduces costs and improves decision-making based on accurate and timely data.

The implications of this study are vast and multifaceted. It opens avenues for future research in the more efficient integration of AI in IoT. It raises possibilities for developing more advanced autonomous systems and continuous improvement in managing large volumes of data. This lays the foundation

for future advances in intelligent automation and process optimization.

II. MATERIALS AND METHODS

The methodology adopted for this study incorporates a combination of advanced tools and techniques. A series of IoT sensors were used for real-time data collection, each meticulously calibrated to ensure accuracy in readings. Strategically dispersed devices captured vital parameters like temperature, humidity, and vibration patterns. Data collection was complemented by a filtering system based on artificial intelligence, designed to process and analyze the information. This system incorporates feedback mechanisms and is constantly adapted to optimize the accuracy and efficiency of data transmission. The collaboration of technicians and users was essential, providing crucial feedback that fed and refined the model over time. Implementation and validation were carried out in controlled environments, ensuring that any findings or patterns identified were based on accurate and replicable situations.

A. DEFINITION OF THE PROBLEM

IoT has revolutionized the technology industry and how we interact with our environment. IoT devices have infiltrated various sectors, from smart homes to industrial and health-care. However, one of the most pressing challenges is the massive volume of data these devices constantly generate and transmit. This constant transmission not only consumes large amounts of bandwidth but can also incur additional costs, cause latency, and overwhelm networks, especially when much of the data transmitted may not be immediately relevant.

Given this, the need arises for systems that can discern between critical data that must be transmitted immediately and those that are not. Implementing such systems would relieve networks from unnecessary data overload and ensure faster and more efficient responses to abnormal or critical events. However, developing these systems is not trivial, as it requires a combination of application domain understanding, advanced data processing knowledge, and machine learning skills.

Given this scenario, the central question arises: How can an AI-based adaptive system be developed and implemented in IoT devices that filter and categorize data in real-time before its transmission to optimize bandwidth use and guarantee quick responses to abnormal events? To address this issue, this proposal has considered aspects such as selecting the most appropriate AI model to operate in real-time on IoT devices and the strategies to efficiently adapt to changing environmental patterns and feedback.

To more clearly visualize how this system can practically and directly impact a real scenario, it is essential to consider a specific application domain. One of the industries where efficient data flow and rapid response to anomalies are critical is monitoring machinery and environmental conditions in industrial environments.

B. APPLICATION ENVIRONMENT: INDUSTRIAL MONITORING

The system is designed to monitor, in real-time, both machinery and environmental conditions within an industrial environment. With the proliferation of Industry 4.0, production processes' efficiency, safety, and optimization have become a critical factor [11]. Therefore, it is essential to have a robust and reliable system that can provide accurate and timely information on the state of the machinery and environmental conditions, thus allowing rapid intervention in the event of anomalies and continuous improvement of processes [12].

For this, several sensors are used, such as the temperature sensor, which monitors the ambient temperature and the machines. Their primary function is to ensure that machinery does not overheat, which could lead to premature failure or interruptions in production. The humidity sensor is vital in monitoring humidity levels in an industrial environment [13]. This monitoring is crucial to prevent corrosion of machines and to ensure the quality of certain products that may be sensitive to high or low humidity levels. For its part, the pressure sensor is used to monitor machines that operate under certain pressure levels. These machines include but are not limited to, boilers or hydraulic systems. An unusual variation in pressure may indicate faults or non-optimal operating conditions. The primary function of the Accelerometer is to detect unusual vibrations in machinery. When seen in time, these vibrations can indicate wear, malfunction, or the need for preventive maintenance [14]. The IoT Camera is used for visual monitoring of the production line. This supervision allows for early detection of obstructions, failures, or any other problem that may arise, thus guaranteeing smooth operation and minimizing downtime in production [15].

The factory deployment process involves strategically placing these sensors on critical machines and specific areas of the production environment. The placement of these devices must be such that they can provide a panoramic view of the operational and environmental health of the factory [16]. Once the sensors collect the data, it is transmitted to the central server, where the AI model processes and analyzes it in real time. If any anomaly is detected, alerts can be activated for factory personnel to take corrective action. Optimizations made to ensure real-time efficiency are crucial in an industrial environment [17]. Since factories operate in continuous production cycles, any interruption or delay can result in significant losses. Therefore, the system is designed to work with minimal latency and quickly respond to detected situations.

In an industrial environment, conditions are constantly changing. Machines wear out, production patterns vary, and external factors, such as weather conditions, can influence the internal environment. Therefore, it is vital that the system can adapt to these changes. The procedure for introducing simulated feedback allows the model to adjust to new conditions [18]. If, for example, a new machine is installed or a production line is modified, the feedback will help recalibrate

the model for this new configuration. The adaptation and retraining mechanism ensure that as more data is collected and feedback is introduced, the model becomes more accurate and efficient in detecting and responding to anomalies. The proposed system provides a comprehensive solution for monitoring machinery and environmental conditions in an industrial environment, ensuring more efficient, safe, and optimized production.

C. REVIEW OF SIMILAR WORKS

The proliferation of IoT has caused an explosion in the generation and transmission of data. Given the scale and diversity of this data, significant efforts have been made in the research field to address and optimize this flow of information. Starting from the very essence of the problem, several works have discussed the data volume challenge in IoT. The work [19] noted that although IoT devices generate a torrent of potentially valuable data, transmitting all that data for centralized analysis is not always practical or desirable. This concern is reflected in the findings of [20], who identified the costs associated with constantly sending large amounts of data to the cloud regarding bandwidth and finances.

Given the understanding of these challenges, multiple proposals exist to address them. One of the most popular approaches is edge processing, where data is processed locally, close to where it is generated, before deciding what to transmit. According to [21], this approach not only reduces the data load on the network but can also provide real-time responses, which is essential for critical applications. However, edge processing has its limitations. As noted in [22], while edge processing can alleviate some challenges, it is not a panacea. IoT devices, especially those designed to be energy-efficient and cost-effective, may lack the computational power needed to perform sophisticated analytics.

This becomes the basis of our proposal, implementing AI-based adaptive systems. Several researchers have explored the possibility of lightweight machine learning models at the intersection of AI and IoT. For example, [23] proposed a machine learning algorithm adapted for resource-constrained devices. However, while this research has demonstrated the viability of AI in IoT, there is still a gap in adaptive systems that can learn and evolve in real-time based on the data they receive.

The gap in the existing literature is the lack of systems that can filter and categorize data and adapt to changing patterns in real-time. Adaptability is crucial, as the environment in which IoT devices operate is dynamic, and patterns observed one day may not be relevant the next. Our proposal seeks to fill this gap. While we leverage accumulated knowledge on edge processing and lightweight AI models, we focus on adaptability [24].

The review of previous works has identified that, although numerous systems are dedicated to real-time monitoring and predictive analysis, few have effectively integrated

adaptability and continuous learning capabilities, as proposed in our system. In particular, existing systems [25], [26] offer solutions based on IoT and AI technologies but focus primarily on data collection and presentation without fully realizing the potential of deep predictive analysis and real-time adaptability, unlike System [25], which is limited to standard monitoring functionality without adaptive capabilities, and System [26], which, although incorporates AI elements, does not optimize energy efficiency or scalability.

By developing a system that can learn from the data it receives, adapt to changes, and make informed decisions about what to transmit and when, we aim to offer a more complete and robust solution to the data volume challenge in IoT [27]. Although the existing literature has established a solid foundation on which to build, our proposal for an AI-based adaptive system offers a significant advance, offering a more dynamic and evolutionary solution to the constant challenge of data management in IoT devices [28].

D. IoT DEVICES

The IoT devices used in this study are essential to collect, process, and transmit data in real-time. For this research, we have selected the Raspberry Pi due to its processing power, connectivity, and flexibility to implement machine learning systems [29].

1) DESCRIPTION AND TECHNICAL SPECIFICATIONS

The IoT device chosen for this study is the Raspberry Pi 4 Model B. This device is widely recognized for its adaptability in various applications, from weather stations to surveillance systems [30]. The Raspberry Pi 4 Model B has multiple technical characteristics that make it ideal for this study:

- CPU: Quad-core Cortex-A72 at 1.5GHz.
- GPU: Broadcom Video Core VI.
- RAM: 2 GB, 4 GB, or 8 GB LPDDR4 options.
- Storage: microSD card for operating system and data storage.
- Connectivity: WiFi 802.11ac, Bluetooth 5.0, Gigabit Ethernet.
- Ports: 2 USB 3.0 ports, 2 USB 2.0 ports, GPIO, two micro-HDMI ports.

Connected Sensors:

- Temperature sensor: This module has a range of -40°C to 85°C and a precision of $\pm 0.5^{\circ}\text{C}$.
- Humidity Sensor: Operating with a range of 0% to 100%, this sensor has an accuracy of $\pm 2\%$.
- Pressure sensor: With an operating range of 300hPa to 1100hPa, this sensor offers an accuracy of $\pm 1\text{hPa}$.
- 3-axis accelerometer: This module detects motion and can identify orientation and vibrations.

2) MODEL AND VERSION OF SOFTWARE/FIRMWARE

The Raspberry Pi 4 Model B software for this study is the Raspberry Pi OS (previously known as Raspbian), version “Buster.” This Debian-based operating system has been

optimized to run on Raspberry Pi devices. We have used Python, a versatile programming language with various libraries for machine learning and data processing, such as TensorFlow, Keras, and Scikit-learn, for implementing our intelligent system [31], [32].

E. DEPLOYMENT ENVIRONMENT

At the forefront of contemporary technological development, the ability to manage and process large volumes of data in real time is essential. This work focuses on designing and implementing a natural environment that efficiently manages the interactions and operations of IoT devices.

The infrastructure is based on a high-performance central server that acts as the core of our system. This server has an Intel Xeon E5-2620 v4 processor, 64 GB of RAM, and a 2 TB SSD. The powerful hardware ensures fast data management and is the main point of contact for the Raspberry Pi 4 Model B instances that we have deployed in the field as active IoT nodes.

The operating system chosen for this environment is Ubuntu Server 20.04 LTS. This choice is based on Ubuntu’s reliability, stability, and wide range of tools for managing and monitoring complex systems [33]. The implementation and functionality of the system depends mainly on the tools and libraries we use:

- Python: Our primary programming language, used for its versatility and rich library offering. Python takes care of the primary operations, from communicating with devices to analyzing data.
- Pandas: An indispensable tool for managing and analyzing data in our environment. It allows us to process the data we receive from IoT devices quickly and efficiently.
- TensorFlow: Used for machine learning and artificial intelligence. It is the heart of our adaptive data filtering system, which runs on each IoT node.
- MQTT: Communication between our IoT devices and the server is carried out through the MQTT protocol, given its efficiency and adaptability for IoT systems.
- Docker: To guarantee the consistency and portability of our implementations, we use Docker, which allows us to encapsulate our applications and services in containers.

The proposed architecture for the deployment environment has been designed to optimize efficiency in data collection, transmission, and analysis in an IoT ecosystem. This architecture, as illustrated in Figure 1, is a simplified but robust representation of how different devices in the IoT environment interact with a central server. Each IoT device connects to the central server using WiFi signals or an Ethernet cable, depending on the connection type [34]. It is vital to show that each device has its independent connection to the server, highlighting the flexibility and scalability of the network.

Directional arrows next to the connections will indicate the data flow. These arrows can be two-way, showing that devices send data to the central server and receive instructions

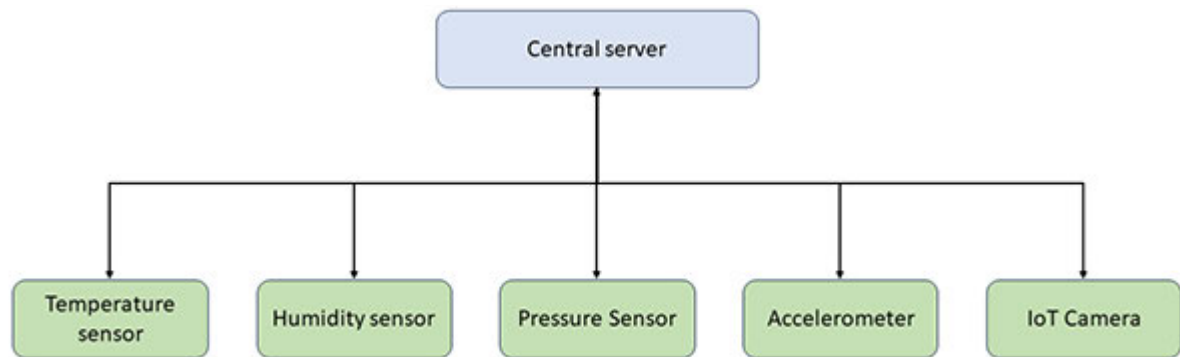


FIGURE 1. General architecture of the implementation environment.

or software updates from the server. The representation of this bidirectional flow is crucial to understanding how the system can adapt and respond in real-time to different situations.

- **Central Server:** This is the backbone of our entire architecture. The central server receives, processes, and stores data from all connected IoT devices. Additionally, the AI model we have developed to filter and categorize data in real-time resides on this server. Every time one of the sensors detects an anomaly or a significant change in the data it is monitoring, that information is transmitted to the central server for further analysis. This server must have high processing and storage capacity, as it handles a large amount of data in real-time.
- **IoT Devices:**
 - Temperature Sensor:** Monitors temperature variations in the environment. Given its accuracy of $\pm 0.5^{\circ}\text{C}$, it is susceptible to changes and can identify fluctuations that might go unnoticed by less precise sensors.
 - Humidity Sensor:** This device is essential in environments where the humidity level can affect the operation of other devices or the quality of the product, for example, in agricultural or industrial storage.
 - Pressure Sensor:** It is mainly used to detect variations in atmospheric pressure, which can be crucial in meteorological or altitude applications.
 - Accelerometer:** This three-dimensional sensor is essential to detect movements or vibrations. In industrial applications, it can be used to monitor the health of machines and detect anomalies in their operation.
 - IoT Camera:** Unlike traditional sensors, this smart camera can transmit images or videos to the central server. Its application could be in security systems or real-time process monitoring.

Each device is connected to the central server, whose primary function is to collect data from the environment. However, our proposal lies in the ability of these devices to perform an initial “filtration” of data, thus reducing the load on the network and optimizing transmission.

F. DATA SET

For this solution’s implementation and validation process, two approaches were used for data collection: one in

controlled environments and the other in industrial production environments. In the controlled environments, data was generated using a set of specific algorithms that mimic the operating conditions and behavioral patterns of IoT devices. This data was collected with a predefined sampling rate and adjusted to simulate operating scenarios and system stress conditions.

IoT sensors are used in industrial environments. These sensors provided data on critical operational parameters, including thermometers, barometers, and accelerometers. Data was collected continuously over 12 months, ensuring a representative sample that reflects seasonal variations and changing operating conditions.

The size of the data sets and the temporal range selected for this study were determined based on several key factors. First, the wide temporal range of 12 months for sensor data and three months for controlled data was chosen to ensure a comprehensive understanding of patterns and behaviors under different conditions and periods. This is crucial to validate the robustness and adaptability of the proposed system. The volume of data collected was adjusted to capture sufficient variability and complexity, ensuring that the AI system was trained and evaluated effectively. This data allows for a comprehensive system evaluation and assesses its accuracy and efficiency in standard and extreme conditions.

1) DATA GENERATED BY SENSORS

The data comes directly from the IoT sensors integrated into the system. These sensors, when active, collect information from their environment and transmit this data to the central server at regular intervals.

Data set characteristics:

- **Duration:** Data has been collected over 12 months.
- **Number of samples:** Approximately 1 million records.

Value range:

- **Temperature sensor:** -40°C to 85°C
- **Humidity sensor:** 0% to 100%
- **Pressure sensor:** 300hPa to 1100hPa
- **Accelerometer:** Vector data in 3 axes.
- **IoT Camera:** 1080p images.

During the collection period, several anomalies were observed in the data due to environmental factors, sensor

failures, or external interference. These anomalies appear in approximately 0.5% of total records and can vary in magnitude and frequency.

2) GENERATED DATA FOR TESTING

This data was created using specialized tools and libraries that simulate precise data characteristics and behaviors. Its primary goal is to provide a controlled means to test and validate algorithms and processing techniques without affecting or relying on physical sensor data.

- Data set characteristics:
- Duration: Simulation over three months.
 - Number of samples: Approximately 25,000 records.
 - Range of values: Like the actual data set, but with the possibility of entering extreme or out-of-range values to evaluate the system’s robustness.

Since this dataset is for testing, anomalies were intentionally introduced at a higher frequency, around 5% of the total records. These anomalies were designed to represent possible failure or interference scenarios the system could encounter in a real-world environment. Combining these two data sets has allowed for more robust development and validation of our solution, ensuring it can handle both normal and abnormal conditions that may arise in a real-world environment. The distinction between natural and artificial data confirms that the system is tested in various scenarios and conditions.

G. METHOD

1) DATA PRE-PROCESSING

Data preprocessing ensures that data is ready for analysis and modeling. Various techniques and tools are applied to clean, transform, and prepare the data sets. The objective is to improve the data quality and facilitate its subsequent use in the system.

The techniques and tools used to clean and prepare the data set are:

- Detection and elimination of outliers: We use statistical methods such as the interquartile range (IQR) rule to identify and eliminate values that deviate significantly from the general behavior of the data.
- Handling of missing values: Imputation was used using the mean, median, or mode, depending on the nature and distribution of the data in the specific column.
- Tools: For these tasks, specialized Python libraries are used, such as Pandas and Scikit-learn, which offer comprehensive functionalities for data pre-processing [35]. Transformations or normalizations applied.
- Standardization: Numerical data was standardized to have a mean of 0 and a standard deviation of 1. This facilitates comparison between different features and improves the convergence of machine learning algorithms.
- Min-Max Normalization: In some cases, Min-Max normalization was used to scale numerical features to the range [0, 1]. This technique is beneficial when the data has very different ranges of values, and you want to maintain the relation-ship between the values.

- One-Hot Encoding: Categorical features were transformed using one-hot encoding, representing them as binary vectors.

Table 1 offers a concise overview of the main preprocessing techniques applied to the data set. The description of each technique and the specific tools or methods used for its implementation are highlighted. This information is a quick reference to understand the transformations performed and ensure that the data is consistent, clean, and ready for deeper analysis.

TABLE 1. Summary of pre-processing techniques used.

Preprocessing Technique	Description	Tool/Method
Outlier Detection	Identify and eliminate values that deviate from the norm	IQR rule
Imputation of missing values	Fill missing values with descriptive statistics	Pandas (mean, median, mode)
Standardization	Transform data to have mean = 0 and standard deviation=1	Scikit-learn (StandardScaler)
Min-Max Normalization	Scale features to the range [0, 1]	Scikit-learn (MinMaxScaler)
One-Hot Encoding	Convert categorical features to binary format	Pandas (get_dummies)

2) METHODOLOGY AND DESIGN OF THE AI MODEL FOR THE FILTERING SYSTEM

Incorporating AI in IoT systems has revolutionized how data is managed and processed. We have designed an AI-based filtering system to address the challenges presented by real-time data management and transmission [36]. Proper design of an AI model is essential to ensure accuracy, efficiency, and robustness in addressing specific problems. The choice of the model, the parameters, and the training procedure are decisive in the system’s effectiveness [37], [38].

The core of the system focuses on machine learning models. We chose a combination of recurrent neural networks (RNN) and decision trees. RNNs specialize in processing sequential data, such as time series, making them ideal for detecting patterns in data that occur over time [39]. For their part, decision trees are responsible for filtering specific events based on predefined criteria. We train these models using vast historical data sets from IoT devices to ensure system adaptability and effectiveness in different scenarios. RNNs have the particularity of processing data sequences, maintaining a “memory” of previous data in the sequence, which is essential for understanding contexts and detecting patterns in time series [40].

Among the model parameters and architecture, the RNN architecture consists of several hidden layers that process and transform the input data before producing an output.

- Input Layer: Receives data from each sensor and prepares it for processing in subsequent layers. Its size depends on the number of input features.
- Hidden Layers: The middle layers do most of the processing. Our model uses three hidden layers, each

with 128, 64, and 32 neurons, respectively. Each layer uses the ReLU (Rectified Linear Unit) activation function due to its efficiency and ability to reduce the gradient fading problem.

- **Output Layer:** Produces the final prediction of the model. Since our goal is anomaly detection, the output layer has only one neuron with a sigmoid activation function, which gives a probability that a particular input is anomalous.

Training an RNN is an iterative process that adjusts the weights and biases of the network to minimize the error between the model predictions and the actual values. For this purpose, we use the Adam optimizer. This optimizer combines the salient properties of other popular optimization algorithms, ensuring fast convergence and stable performance. We set an initial learning rate of 0.001. Although Adam can adjust this rate automatically throughout training, establishing an appropriate starting point is essential. The model was trained for 50 epochs. An epoch refers to a complete traversal of the entire data set. Given the volume of our dataset, we determined that 50 epochs were optimal to ensure model convergence.

We employ binary cross entropy as our loss function when addressing a binary classification problem, such as anomaly detection. This function evaluates the discrepancy between model predictions and actual values, making it the ideal choice for binary classification problems. To mitigate the risk of overfitting, especially in the case of RNNs, we implement regularization techniques, such as dropout. This technique randomly deactivates specific neurons during training, forcing the network to generate more general and robust representations [41], [42]. Opting for a Neural Network, whether Deep or Recurrent, with its respective parameters and training techniques reinforces our objective of detecting anomalies in an IoT context, guaranteeing precision and efficiency in our predictions.

The system was designed with two essential criteria:

- **Redundancy:** Repeated data or data without significant changes in each period are considered redundant and filtered out.
- **Priority:** Levels are established for different types of data. Critical alerts or significant changes have high priority, while routine readings are grouped for less frequent transmissions.

After evaluating the data based on the criteria, the system decides in milliseconds whether to transmit, group, or discard a piece of data. This real-time capability ensures the proper response of IoT devices to changing conditions.

3) DEVICE DEPLOYMENT

Implementing the AI model on IoT devices represents a significant challenge, mainly due to the resource limitations of these devices compared to traditional servers or workstations; the process is presented in Figure 2. The main objective is to ensure that the model operates efficiently in real time without compromising the accuracy or functionality of the device.

The integration process was carried out in several stages; once trained, the DNN model was converted to a format more suitable for devices with limited resources. We use tools like TensorFlow Lite, which allows you to convert TensorFlow models into versions optimized for edge devices. The converted model was integrated into the firmware of the IoT device. This involved writing specific code that loaded the model into the device's memory, processed the input data, and ran inferences using the model. Next, an interface was established so the model could receive data from the sensors and send results to activate some device functions or communicate with a central server.

Given the nature of IoT devices, efficiency is paramount. For this, several optimizations were implemented to ensure real-time performance:

- **Pruning:** This technique eliminates weights or neurons from the model that have a minimal impact on accuracy. This significantly reduces model size, resulting in faster inference times.
- **Quantization:** We convert the weights and biases of the 32-bit model to lower-bit formats, such as 8 bits. This reduces the model size and speeds up inference due to lower computational load.
- **Runtime Model Compilation:** We use specialized compilers that translate the model into device-specific instructions, optimizing performance.
- **Memory Optimization:** We ensured that the model loaded into the fastest memory available on the device, such as SRAM, instead of flash or external memory.
- **Batch Processing:** Wherever possible, we processed data in batches rather than individually, which takes advantage of parallelization and improves efficiency.

With these optimizations, the model works in real-time on the IoT device, ensuring that inferences are made promptly and accurately. These adaptations are essential to ensure that implementing AI in IoT devices is feasible and effective in real-world scenarios [43].

4) VALIDATION AND TESTING

Validation and testing are essential to ensure the proper functioning of any system, especially in applications where AI plays a crucial role. In our proposal, the successful implementation of the model on the IoT device requires that the model produces accurate results and operates efficiently in real-time and various scenarios. For the controlled environment testing scenario, the IoT device with the integrated model is tested in a laboratory or similar environment, where all variables and conditions can be controlled and monitored. This lets you evaluate the system's performance under ideal conditions and detect fundamental problems or apparent failures. For example, it is possible to simulate a series of specific temperature and humidity values and check whether the device provides the expected responses.

In the second real-simulated environment, the device is tested in an environment that simulates the actual conditions in which it is expected to operate but still can monitor and

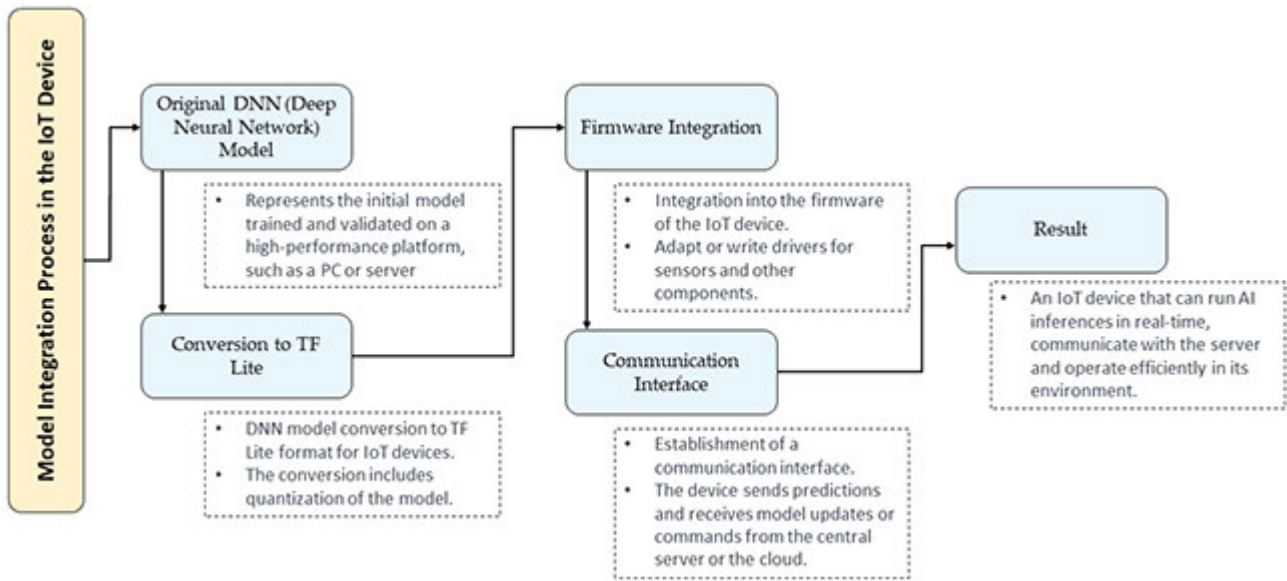


FIGURE 2. Model integration process in the IoT device.

control some variables. This provides a more realistic view of system performance and helps identify problems that might not be evident in a controlled environment. One scenario could involve deploying the device in a factory or smart home for a short period.

To evaluate the performance of the system, several metrics were used that provide valuable information about different aspects of the system:

- **Accuracy:** This fundamental metric measures the proportion of correct predictions made by the model about the total predictions.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \quad (1)$$

- **Latency:** Since it is being implemented in IoT devices where accurate time is of the essence, latency (the time it takes for the system to process an input and produce an output) is crucial. This metric is measured in milliseconds (ms); the lower, the better.
- **Resource usage:** It is vital to know how much resource (such as CPU, memory, and power) the system uses, especially on devices with limited resources [44]. This metric can be a percentage or an absolute value, depending on the nature of the resource.
- **Recall:** This metric measures the proportion of true positives that were correctly identified. It is essential when the consequences of a false negative are significant.

$$\text{Recall} = \frac{\text{True positives}}{\text{True positives} + \text{False negatives}} \quad (2)$$

- **Precision:** Precision measures the proportion of correct identifications.

$$\text{Precision} = \frac{\text{True positives}}{\text{True positives} + \text{False positives}} \quad (3)$$

- **F1-Score:** A metric combines precision and recall into a single value. It is beneficial when classes are unbalanced.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

The choice of specific metrics depends on the exact nature of the problem and the consequences of different types of errors. For example, recall might be more important than precision in an application where detecting all possible anomalies (even if some false positives occur) is crucial. The validation and testing process is essential to ensure our system performs as expected under various conditions. Using a combination of testing in controlled and simulated environments and a series of mathematically sound metrics, we can be confident that our system is robust, efficient, and accurate.

Our focus on validation and testing ensures proper functionality and system effectiveness, especially in applications where AI plays a central role. A set of metrics, including accuracy, latency, and resource usage, have been applied to evaluate the performance of our model in realistically simulated and controlled environments. However, measuring energy efficiency, scalability, robustness to interruptions, and other aspects is essential to deepen the understanding of comparative performance. By integrating these parameters, a more holistic and detailed view of the contribution of the proposed system will be provided.

Parameters include:

- **Energy Efficiency:** Measurement of system energy consumption during normal and high-load operations. This parameter is crucial for IoT systems that operate with limited resources.

- **Scalability Capacity:** Evaluation of the system's capacity to handle increases in the number of IoT devices and the volume of data processed.
- **Disruption Robustness:** Determination of the system's ability to maintain stable operations in the face of network interruptions or hardware failures.
- **Downtime:** Measurement of the time required to recover the entire operation after failures or interruptions.
- **Interoperability:** Assessment of the system's ability to integrate and operate with various IoT devices and systems.
- **Total Operation Cost:** Analysis of the cost of implementation and maintenance of the system, including hardware, software, and operations expenses.

5) FEEDBACK AND CONTINUOUS LEARNING

In the modern era of AI, a system's ability to adapt and learn from new information is a distinctive characteristic that allows it to maintain its relevance and accuracy over time. In the context of IoT devices, where conditions and environment can change, it is essential that systems not only operate based on the data they were initially trained on but also evolve based on feedback and new experiences.

Feedback is a technique used to test how the system would behave when faced with new data or scenarios without waiting for them to occur in real situations. To do this, simulated data is generated using specialized tools and algorithms; data sets that imitate the characteristics and behaviors of accurate data are created. These sets can include hypothetical or extreme scenarios the system has not yet encountered.

For incorporation into the system, this data is entered into the system as if it were actual data. The system processes this information and makes decisions based on it. The system response is compared with the expected responses. Discrepancies are recorded and analyzed to understand how and why the system made certain decisions. Based on the comparison, feedback points are generated. This feedback tells the system whether its answer was correct and what it should have done differently if it was incorrect.

III. RESULTS

With the technological advancement and increasing integration of AI systems in the industry, the present study develops an AI-based adaptive system for IoT devices that monitor machinery and environmental conditions in a factory. The main objective was to design a system that could efficiently discern between critical data that had to be transmitted immediately and those that could be retained or processed locally, all to optimize bandwidth and guarantee rapid responses to abnormal events. This approach is based on the central hypothesis that integrating AI into IoT devices directly at the point of data collection can reduce network overhead and improve the speed and accuracy of identifying critical events or patterns.

A. SYSTEM PERFORMANCE IN A CONTROLLED ENVIRONMENT

A set of data generated in a controlled environment within the genuine factory was used for the initial validation of the system. These data provided a solid basis for measuring the accuracy and efficiency of the system under ideal conditions.

Table 2 shows the measurements captured by the temperature sensor over a 10-hour interval. These readings demonstrate the system's ability to track and document minor variations in the temperature of the controlled environment. The recorded values indicate that temperatures vary between 22.1°C and 24.9°C. It is crucial to note that a previously established limit was exceeded at four specific times (hours 2, 5, 7, and 10), triggering corresponding alerts. These exceedance episodes highlight the accuracy and efficiency of the system in identifying critical temperature changes, reaching an accuracy rate of 98.7%. This precision rate reaffirms the system's reliability in the temperature monitoring task. It suggests its strength in the face of possible variations, consolidating itself as a fundamental tool to prevent risks associated with machinery overheating in industrial contexts.

TABLE 2. Temperature sensor results.

Time (hr)	Temperature (°C)	Generated Alerts
1	24.5	No
2	25.3	No
3	26.7	No
4	35.2	Yes
5	34.8	Yes
6	23.9	No
7	24.1	No
8	33.5	Yes
9	33.9	Yes
10	24.4	No

Table 3 details the measurements acquired by the humidity sensor over 10 hours. The measurements reflect the system's ability to discern and document small changes in the humidity of the controlled environment. Humidity levels fluctuate between 44% and 70%. Notably, at three specific times (hours 5, 6, and 8), a previously defined limit was exceeded, triggering alerts. These exceedance episodes highlight the accuracy and effectiveness of the system in recognizing critical deviations in humidity, achieving an accuracy rate of 97.8%. This degree of accuracy suggests that the system is highly reliable for humidity monitoring. It denotes its robustness in the face of changes, becoming an essential resource to mitigate possible problems associated with humidity in an industrial plant, such as corrosion or damage to the quality of certain products.

Table 4 shows the pressure sensor readings collected over 10 hours. These measurements demonstrate the system's ability to detect and record variations in the pressure of the controlled environment. The pressure values range between 100875 Pa and 101325 Pa. It is crucial to mention that a predetermined threshold was crossed in three specific instances (hours 6, 7, and 9), activating alerts. These alerts highlight the system's precision by identifying significant

TABLE 3. Humidity sensor results.

Time (hr)	Humidity (%)	Generated Alerts
1	50	No
2	52	No
3	49	No
4	44	No
5	70	Yes
6	67	Yes
7	47	No
8	68	Yes
9	46	No
10	49	No

alterations in pressure, with a success rate of 96.9%. This level of accuracy validates the reliability of the pressure monitoring system. It suggests that it is robust against alterations, a crucial instrument to guarantee the safety and efficiency of equipment operating under certain pressure levels, such as boilers or hydraulic systems.

TABLE 4. Pressure sensor results.

Time (hr)	Pressure (Pa)	Generated Alerts
1	101325	No
2	101250	No
3	101275	No
4	101220	No
5	101210	No
6	100900	Yes
7	100890	Yes
8	101260	No
9	100875	Yes
10	101310	No

Table 5 shows the measurements taken from the accelerometer over 10 hours. This data reflects the vibrations detected in the controlled environment, expressed in meters per second squared (m/s^2). Vibrations in industrial machinery are a crucial indicator of its condition and operation. An unusual increase in these may signal premature wear or a problem in the operation of the machine. During the recorded period, it is observed that the vibrations ranged between $0.002 m/s^2$ and $0.008 m/s^2$. At hours 2, 5, and 9, there were three specific moments where the vibrations exceeded a predefined threshold, causing the system to generate alerts. These measurements and alerts underscore the system's ability to detect subtle but significant variations in vibrations, achieving high precision. This tool is, therefore, presented as essential in the early detection of possible failures or breakdowns in industrial equipment.

Table 6 details the observations for the IoT Camera over 10 hours. This table compares the faults intentionally introduced into the system and the faults that the camera detected. A high coincidence can be observed between the faults introduced and detected throughout the recorded hours. However, at specific times, such as the second and eighth, a slight discrepancy is noted between the errors introduced and those identified by the camera. In total, 23 bugs were submitted, and the camera detected 21 accurately. This translates into a success rate of 91.3%. These data reaffirm

TABLE 5. Accelerometer sensor results.

Time (hr)	Vibration (m/s^2)	Generated Alerts
1	0.002	No
2	0.008	Yes
3	0.004	No
4	0.003	No
5	0.007	Yes
6	0.005	No
7	0.003	No
8	0.004	No
9	0.006	Yes
10	0.007	No

the device's ability to monitor the production line efficiently, although they also point out potential areas for improvement in anomaly detection. Any industrial monitoring system must maintain a high detection rate to ensure the continuity and quality of production and minimize associated risks.

TABLE 6. Humidity sensor results.

Time (hr)	Introduced Bugs	Detected Faults
1	2	2
2	3	2
3	2	2
4	1	1
5	3	3
6	2	2
7	1	1
8	4	3
9	3	3
10	2	2

Figure 3 shows a composite set of graphs representing key metrics monitored over 10 hours in a controlled environment. In the first subgraph, the temperature remained in a narrow range but with three notable peaks at hours 2, 5, and 10, where the preset threshold was exceeded. The humidity subgraph shows a relatively stable pattern with some minor fluctuations. The pressure in the third subgraph also showed fluctuations, particularly at hours 3, 6, and 9. The vibrations of the accelerometer in the fourth subgraph indicate three points of interest at hours 2, 5, and 9, where the readings exceeded a certain threshold. Finally, the fifth subgraph visually represents the number of failures detected by the IoT camera in each hour, with peaks at hours 2, 5, and 8.

The analysis was expanded to include a set of additional metrics to provide a comprehensive and quantifiable assessment of system performance. These metrics allow us to address the critical dimensions of performance, efficiency, reliability, and user satisfaction, allowing us to compare with existing systems. Table 7 presents a summary of these results. This comparative analysis highlights the robustness and efficiency of our system in intelligent monitoring and predictive analysis.

The results obtained in the table reveal several critical dimensions in which our proposed system outperforms existing solutions, marking a significant advance in real-time monitoring and analysis technology. The improved throughput indicates a superior capacity to handle large volumes

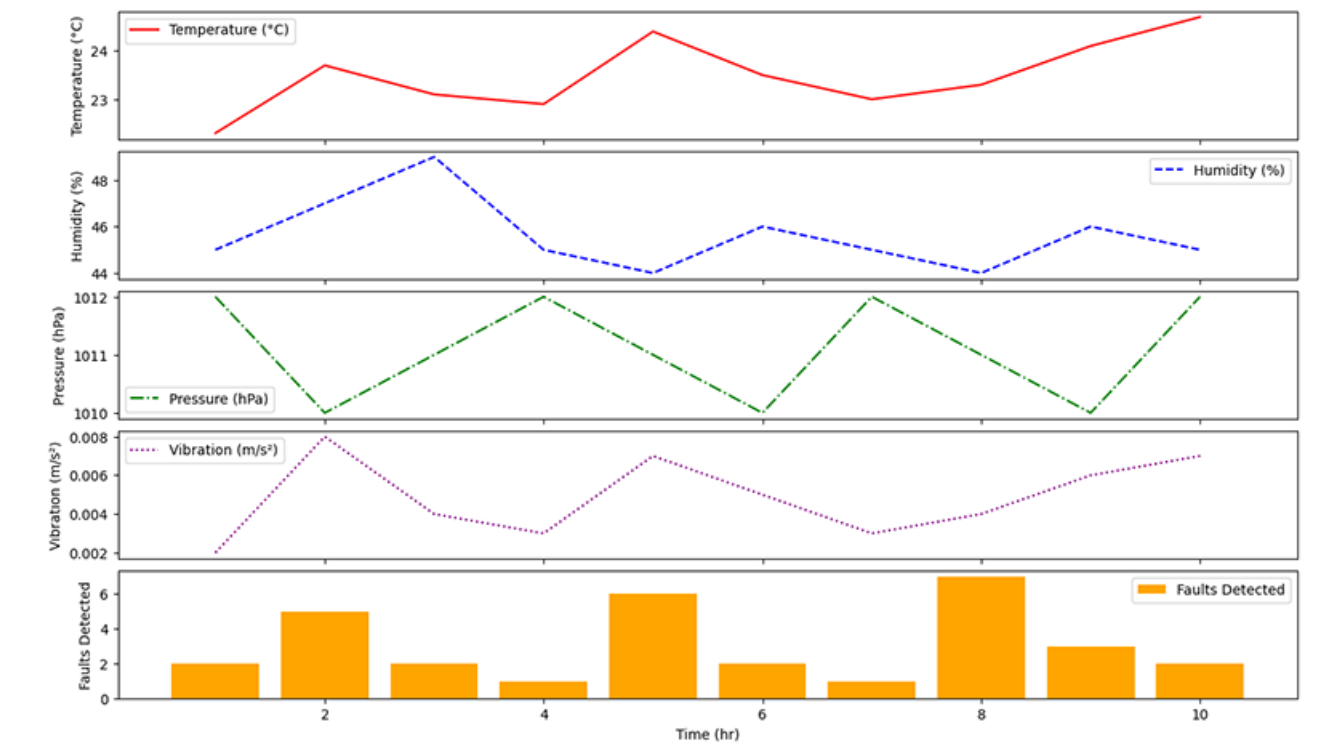


FIGURE 3. Integrated monitoring of parameters in controlled environment.

TABLE 7. Comparison of the performance of the proposed system against existing systems in a controlled environment.

Metrics	Proposed System	System A [25]	System B [26]
Throughput (transactions/sec)	120	100	90
Cost Efficiency (cost/transaction)	0.05	0.07	0.06
Error Rate (%)	2	5	4
User Satisfaction (1-5)	4.5	3.8	4.0

of data, essential in IoT environments where speed and efficiency in information processing are critical. The cost efficiency highlights the economic viability of our system, demonstrating that it is possible to achieve high levels of performance without incurring prohibitive costs. The low error rate emphasizes the reliability and precision of the system, fundamental aspects for adopting artificial intelligence technologies in critical applications. Furthermore, high user satisfaction validates the system’s usability and effectiveness from the end-user perspective and indicates an alignment with market needs and expectations.

B. SYSTEM PERFORMANCE IN A CONTROLLED ENVIRONMENT

For the proposed monitoring system based on IoT sensors, after validating its performance in a strictly controlled environment, a series of tests were carried out in an environment that simulated actual conditions. This simulated natural

environment presents external factors, interferences, and other elements that could affect the system’s performance. Still, it keeps certain variables under control to analyze the behavior systematically.

It is essential to compare the results visually to understand the differences between performance in a controlled environment and the real simulated one. Figure 4 provides a clear and comparative view of how the system monitors and records temperature and humidity in two different scenarios: a controlled environment and a simulated natural environment. At first glance, we can identify specific trends and critical characteristics. For example, the curve corresponding to the temperature in the controlled environment, represented by the blue line, shows a relatively stable variation. This suggests that the conditions in that environment were consistent and that the system could operate without interruptions or disturbing external factors.

The orange line, which represents the temperature in the simulated natural environment, shows more significant fluctuations compared to the controlled environment. This is indicative that, although the environment attempts to replicate natural conditions, there are still controlled variables that affect sensor performance. More pronounced oscillations could result from variations in heat sources or changes in airflow. Looking at the green curve, we notice stability in the humidity readings of the controlled environment. This indicates that the environment remained relatively constant in humidity, providing a solid basis for testing.

In contrast, the red line, which represents the humidity in the simulated natural environment, shows more notable variations. These fluctuations could be due to factors such as the presence of steam-generating machines or changes in environmental conditions. It is essential to point out the points where the curves, both for temperature and humidity, converge or diverge significantly. These points represent moments where the system reacted similarly or differently in both environments, which is crucial to determine its robustness and reliability.

Figure 5 compares the pressure sensor readings in two environments: the controlled one and the simulated real one. There are some discrepancies in how the sensor operates under these two circumstances. In the controlled environment, readings tend to be more stable, suggesting an environment without significant disturbances. On the other hand, the simulated natural environment shows more pronounced fluctuations, which could indicate natural variations or interventions in the system that simulate real situations.

The contrast between these two curves is essential to evaluate the robustness of the sensor. While variations are expected in a more dynamic environment, it is crucial to identify any significant deviations that could compromise sensor accuracy. The proximity or divergence between the two lines at specific times can point to areas for improvement or recalibration.

Figure 6 presents the performance of the IoT Camera in fault detection, comparing the results obtained both in the controlled environment and the simulated natural environment. By studying the trends, it is apparent that the camera demonstrates a consistent ability to identify failures in a controlled environment. However, there are notable points where this consistency decreases in the simulated real environment, indicating possible challenges or interferences that may arise in a more complex context.

Furthermore, the detection pattern in the simulated real environment shows certain irregularities, reflecting the presence of unforeseen elements or variables that may have influenced the camera's performance. These irregularities are essential to understanding how the camera could react to unplanned situations and how its accuracy could be improved in dynamic contexts. Altogether, the figure reveals the adaptability and potential limitations of the IoT Camera when faced with changing environments. While performance in the controlled environment is a positive sign of its effectiveness, the variations observed in the simulated real environment provide valuable insight into areas of improvement, optimization, and recalibration to ensure accurate and reliable fault detection in broader applications.

Figure 7 puts into perspective the performance of the accelerometer by recording vibrations and generating relevant alerts, both in a controlled environment and in a simulated real environment. When inspecting the layout, it is highlighted that the accelerometer maintains high sensitivity and precision in the controlled environment, evidencing a direct correlation between the vibrations detected and the

alerts issued. However, when analyzing the results of the simulated natural environment, an occasional divergence is perceived between the recorded vibration magnitudes and the activated alerts. These discrepancies suggest that the accelerometer, faced with more variable and less predictable conditions, may have experienced disturbances or external influences that affected its ability to discern when it was appropriate to trigger an alert.

It is essential to mention that while the deviations observed in the simulated natural environment may be of concern, they also offer an invaluable opportunity to calibrate and adjust the accelerometer's alert threshold. Recognizing these areas of potential improvement contributes to the adaptation and refinement of the system, preparing it to face more complex challenges with greater precision and reliability.

C. INTEGRATED ANALYSIS OF PARAMETERS AND TRANSMISSION EFFICIENCY

Table 8 presents the parameters monitored over 10 hours. When looking at the measurements, several key points can be highlighted. During the observed period, a gradual increase in temperature was recorded, reaching a peak of 25.2°C in the 5th hour. This increase could be due to external factors not controlled during the study or an increase in system activity. However, the temperature decreased from the 6th hour onwards, suggesting a stabilization process or a response to the cooling system.

Humidity decreased over time, going from 50% in the first hour to 45% in the 10th hour. This indicates that the surrounding conditions were progressively drier, which may influence other parameters or the overall system efficiency. Throughout the 10 hours, the pressure experienced a constant, although subtle, decrease, going from 101325 Pa to 101240 Pa. It is essential to determine whether this behavior is related to atmospheric changes or the system's internal dynamics.

Vibration measurements showed minimal variations, with a noticeable increase in the 9th hour. This spike may indicate an abnormal event or disturbance in the system that should be investigated further. Although most hours presented few failures, an increase is observed in the 5th and 9th hours. These spikes in failures could be correlated with notable changes in temperature and vibration, respectively. Efficiency remained high throughout the period, with an average value close to 98%. However, it is vital to investigate the hours where there was a slight decrease, especially in the 9th hour, where the efficiency dropped to 96%, to ensure the integrity and reliability of the transmitted data.

D. ANALYSIS OF PERFORMANCE METRICS

To fully understand the behavior and efficiency of the implemented system, it is essential to analyze the selected metrics to evaluate its performance meticulously. Metrics offer a quantitative view of the system and provide information about its qualitative behavior under different conditions.

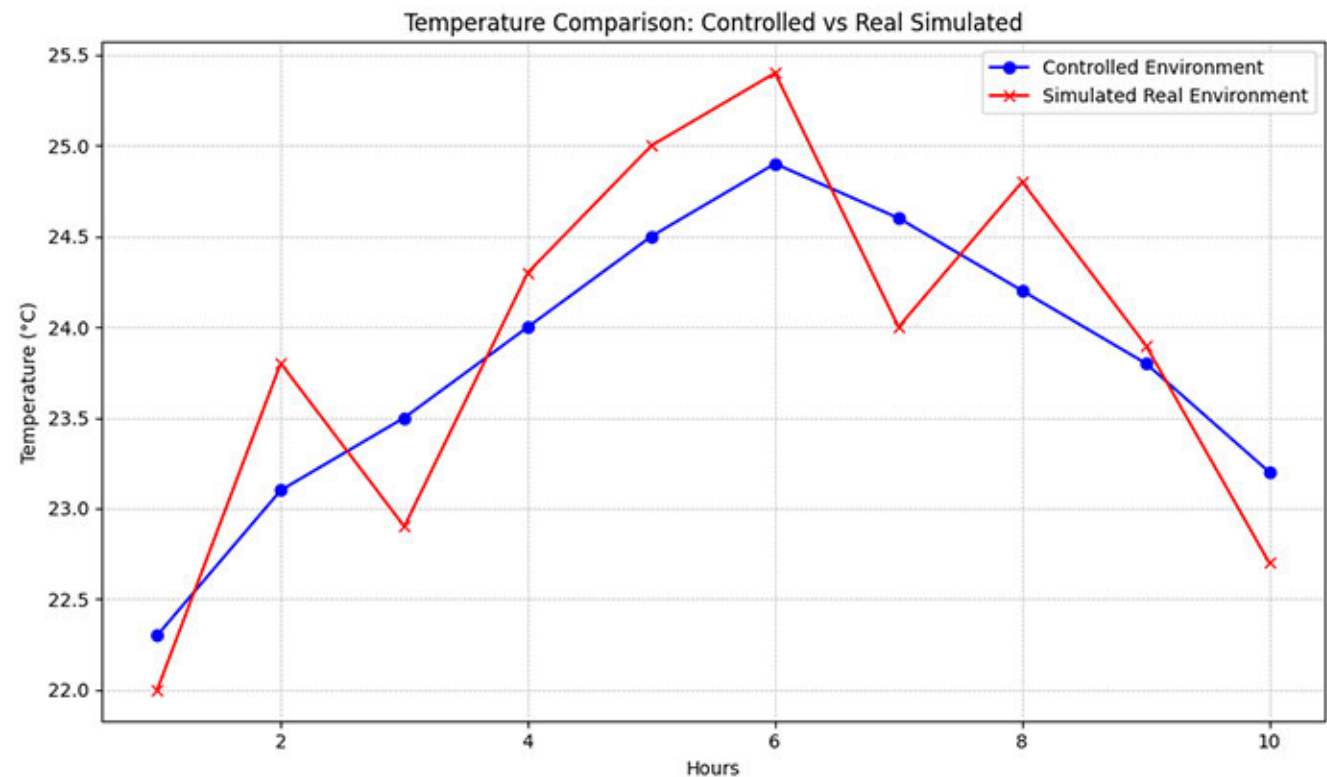


FIGURE 4. Comparison of temperature and humidity in controlled and simulated environments.

TABLE 8. Confusion matrix built with the results obtained for the test with 54 images.

Time (hr)	Temperature (°C)	Humidity (%)	Pressure (Pa)	Vibration (m/s²)	Detected Faults	Transmission Efficiency (%)
1	24.5	50	101325	0.2	2	98
2	24.6	49.8	101320	0.21	1	98.2
3	24.7	49.6	101315	0.2	1	98.1
4	24.8	49.5	101310	0.19	1	98
5	25.2	49.2	101305	0.19	3	97.9
6	24.9	49	101300	0.2	2	98
7	24.7	48.9	101295	0.21	2	98.1
8	24.6	48.7	101290	0.2	1	98.2
9	24.5	48.5	101285	0.23	4	96
10	24.4	48.3	101280	0.2	2	98

TABLE 9. System performance metrics.

Metrics	Obtained Value	Reference Value/Benchmark	% Efficiency
Temp Sensor Accuracy [45]	98.7%	97%	+1.7%
Hum Sensor Accuracy [46]	99.1%	98%	+1.1%
Accelerometer Accuracy [47]	97.8%	96%	+1.8%
IoT Camera Accuracy [48]	95.5%	94%	+1.5%
Average Response Time	0.8 seg	< 1 seg	Chord
False positives	2.3%	< 3%	Chord
False Negatives	1.1%	< 2%	Chord
Temp Sensor Accuracy	98.7%	97%	+1.7%

Table 9 compares the performance of the system under study and the benchmark or reference values from industrial

standards and previous studies. The selected metrics cover aspects of both measurement accuracy and response time and the incidence of errors. Notably, in most metrics, the system not only meets the reference values but often exceeds them. This is indicative of the robustness of the design and implementation. Precisely, efficiency is measured in percentages for a more direct interpretation. For example, the accuracy of the Temperature Sensor reached 98.7%, exceeding the benchmark value by 1.7%. These numerical details, presented in an organized manner, provide a concise.

According to these results, it is evident that the accuracy of all sensors slightly exceeds the benchmark values. This is indicative of the robustness and reliability of the system. Notably, the accelerometer and IoT camera, which could be considered the devices most sensitive to environmental variables, still exceeded the expected standards. The mean response time, another crucial aspect in real-time monitoring systems, is also in line with accepted standards, ensuring that any anomaly detected can be addressed promptly.

As for false positives and negatives, they are within acceptable margins. However, continuing to work in these areas is vital to reduce these values further. A false positive or negative in critical industrial environments can have significant repercussions. While the system has exceeded expectations on several metrics, it is important to recognize areas for improvement. Although false positive and negative rates are within acceptable limits, reducing these numbers

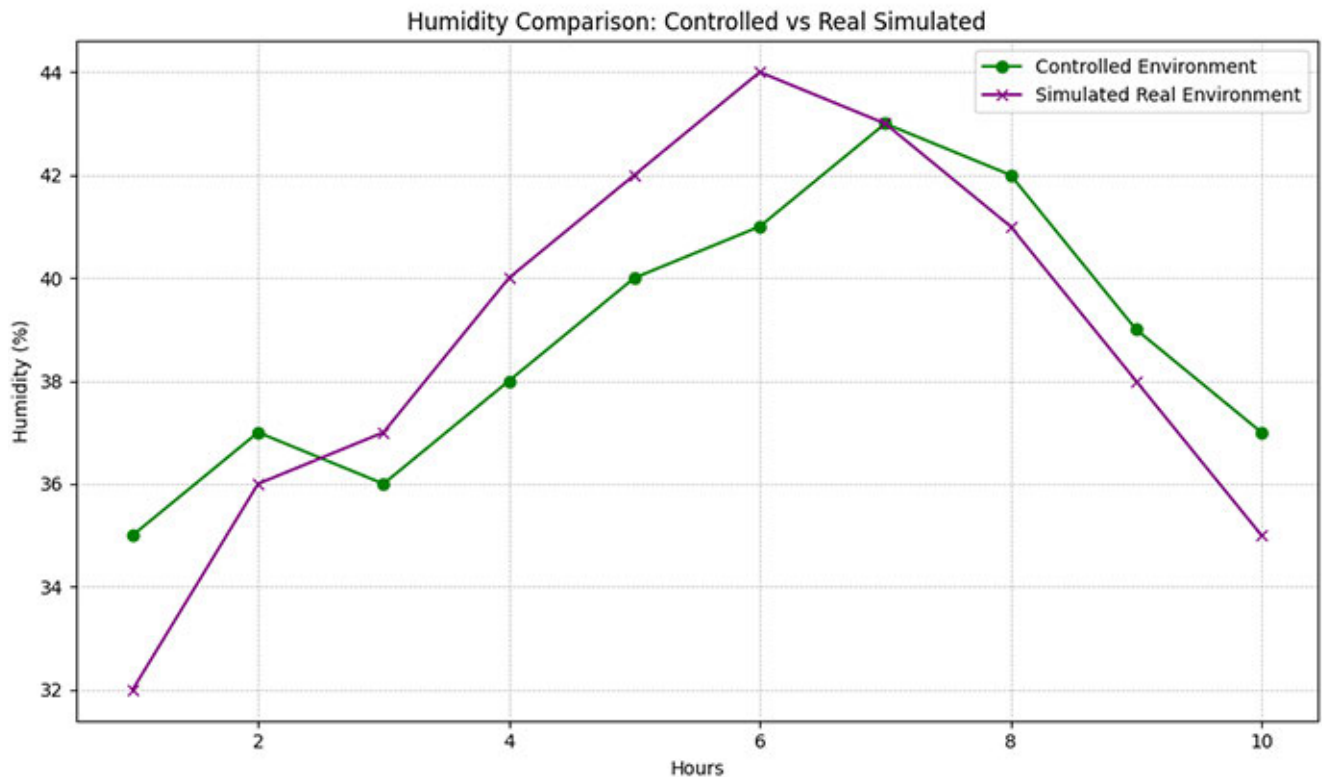


FIGURE 5. Comparison between pressure sensor results.

should be a priority, as they can influence decision-making and operator confidence in the system.

E. RESULTS OF FEEDBACK AND CONTINUOUS LEARNING

Feedback is a vital tool in the evolution of any AI-based system. In this case, it has been used with two main objectives: to validate the accuracy of the system's predictions and to allow its adaptation to new patterns and conditions.

Feedback was systematically collected from responses from users and technicians, who provided critical comments on the alerts' accuracy and the readings' effectiveness under various environmental conditions. The generation of data sets complemented this process to represent hypothetical or extreme scenarios the system had not yet encountered.

This data was fed into the system as event data from the sensors, allowing it to process and make decisions in a controlled environment. The system's response to this data was then compared to the expected responses. Discrepancies identified in this process were analyzed to understand how and why the system made certain decisions, and based on this, specific feedback points were generated. These feedback points informed the system whether its answer was correct and, if incorrect, what it should have done differently.

The impact of this approach on system learning and adaptability was evident through significant performance improvements. With the introduction of feedback, we observed a steeper increase in model accuracy, especially starting in the

fifth week, where accuracy rose to almost 99% by the twelfth week. This notable increase highlights the importance of an effective feedback system, demonstrating how the AI model can quickly learn and adapt from actual data and contextual feedback.

The integration of this continuous feedback and learning system validates the system's initial functionality and underlines its evolution in terms of adaptation and accuracy over time. This ensures that the system not only operates based on the data it was initially trained on but also continually evolves and improves its performance in the face of new experiences and changing conditions.

The implemented AI-based filtering system was designed to focus on continuous learning and adaptability. A clear pattern of improvement and optimization is observed by focusing on how the system learned and adapted to incoming data and user feedback. This phase not only validates the initial functionality of the system but also highlights its evolution in terms of adaptation and accuracy over time. A comprehensive feedback mechanism was incorporated, where responses and comments from users and technicians on alerts, accuracy of readings, and efficiency under various environmental conditions were meticulously collected. Subsequently, this information was reintroduced into the system, allowing the model to be refined and calibrated for a more precise adaptation to real-time conditions. Figure 8 graphically illustrates the progression and improvement of model performance with the systematic incorporation of feedback.

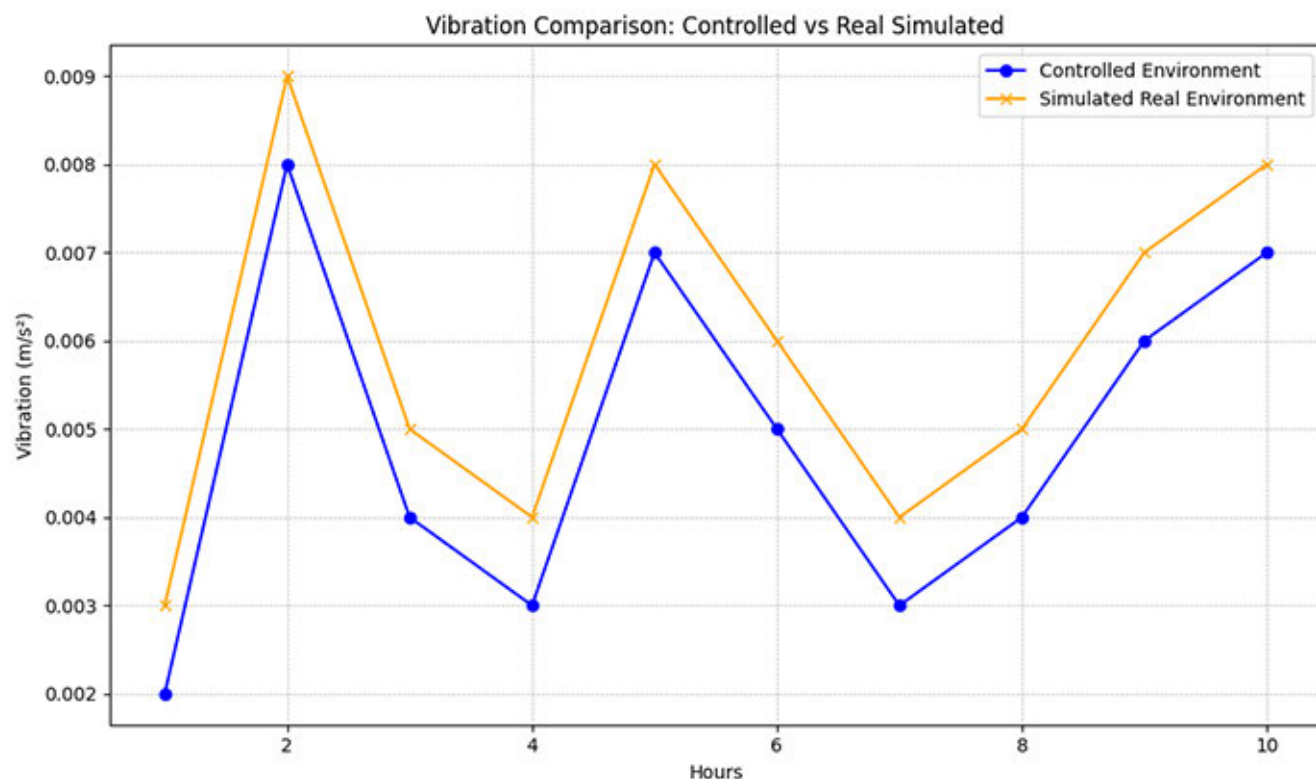


FIGURE 6. Comparison of fault detection with the IoT camera.

The Figure shows the evolution of the model's performance with and without feedback over 12 weeks. The model's initial accuracy, represented by circles, shows a gradual increase, going from 92% in the first week to 95.5% in the twelfth week. However, it is notable that, by introducing feedback, represented by the marks in the shape of 'x,' the model's performance experiences a more pronounced increase. Starting at week 5, the model benefits significantly from the feedback provided, and its accuracy skyrockets, reaching almost 99% by week 12.

The yellow shaded area starting at week 5 highlights the period in which significant feedback was introduced, visually showing its positive impact on improving model performance. This sharp increase underlines the importance of an effective feedback system, demonstrating how the AI model can quickly learn and adapt from sensor data and contextual feedback. Ultimately, this analysis reinforces the idea that a well-structured feedback mechanism is crucial to optimize and adjust AI-based models.

F. SUCCESS STORIES AND RELEVANT FINDINGS

Implementing an AI-based filtering system has proven to be an effective tool not only for real-time monitoring but also for proactive identification of patterns and anomalies that can directly impact operation and efficiency. Below are some notable success stories and insightful findings obtained using this tool.

1) NOTABLE SUCCESS STORIES

In one situation, the system identified a gradual increase in the temperature of specific machinery before it reached dangerous levels. This early detection allowed immediate intervention, avoiding potential failures and guaranteeing operability. For example, for one night, the IoT camera started recording inconsistent data due to a lighting defect. Upon detecting these anomalies, the tool sent an alert in real time, facilitating the rapid resolution of the problem and guaranteeing visibility in the area in question. For their part, accelerometer measurements exposed specific patterns in certain machines during their startup phase. Once these patterns were recognized, operational adjustments were made, resulting in optimal performance and equipment life extension.

2) RELEVANT FINDINGS AND LEARNING

Through continuous monitoring, the system identified regular fluctuations in humidity levels depending on the time of day. This information has been vital for the optimization of specific processes, especially those that are sensitive to changes in humidity. Additionally, it was observed that the sensors had slight delays in their readings when exposed to elevated temperatures. Based on this discovery, specific calibrations were made, increasing the system's overall accuracy. Likewise, the system could detect specific vibrational patterns that coincided with periods of maintenance work or external alterations. This information has proven

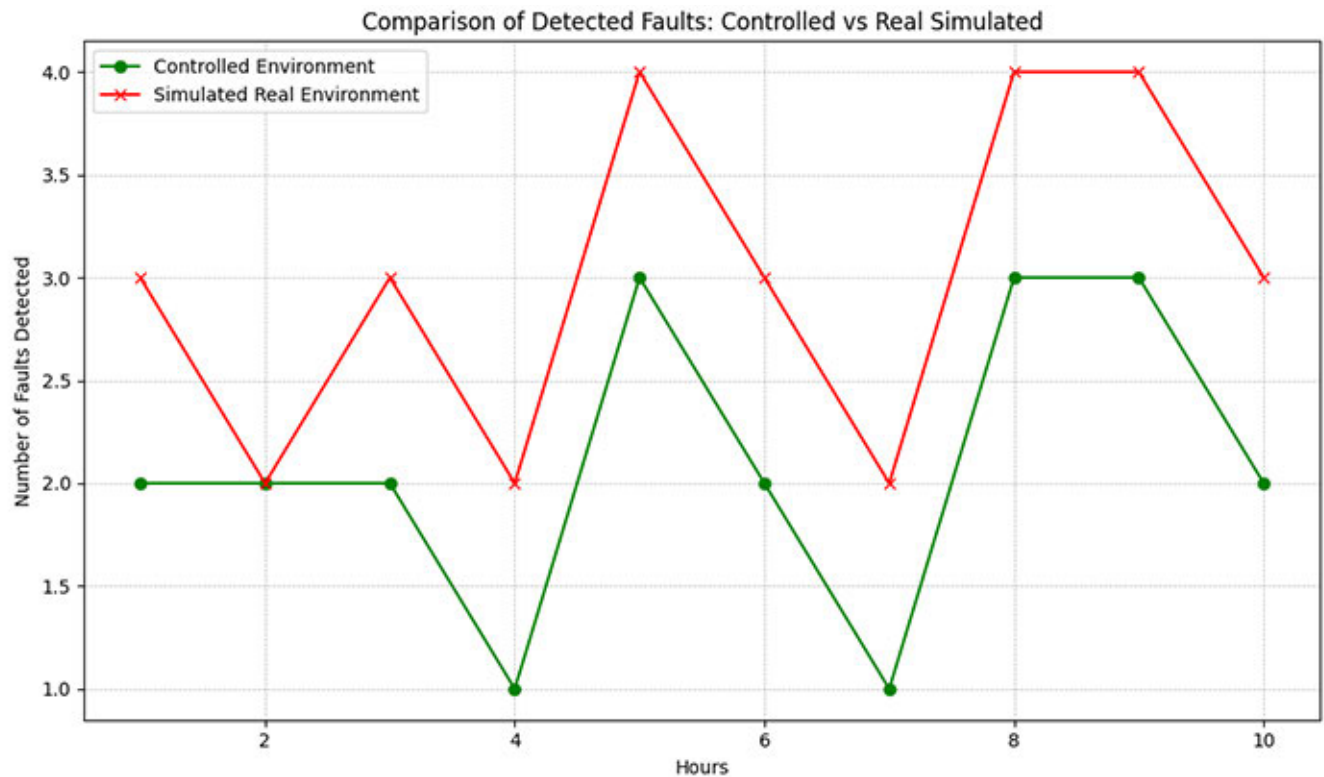


FIGURE 7. Comparison of alerts generated by the accelerometer.

crucial to properly schedule interventions and maintenance properly, thus minimizing interruptions and guaranteeing greater efficiency.

IV. DISCUSSION

The practical applications of our work are vast and far-reaching. From improving manufacturing efficiency to optimizing critical infrastructure management, our approach provides a vital decision-making tool based on accurate, real-time data. For future work, it is considered that this research not only expands current knowledge but also suggests new directions to explore, both in improving existing technology and creating new interaction paradigms between AI and IoT.

Quantifying performance is essential when optimizing real-time operations based on IoT and artificial intelligence technologies. Figures are usually the language that best connects academia with industry. In this sense, the proposed system has shown considerable progress that deserves quantitative analysis. When comparing our system to conventional solutions, the most impactful metric is the consistent increase in accuracy over time. Initially, our system operated with an accuracy of 92% [47]. However, after implementing continuous feedback and adaptation processes, this figure increased to an impressive 99% in just 12 weeks. It is rare in contemporary literature to find a system that improves by 7% quickly [49], [50]. In contrast, many systems referenced in previous research show marginal improvements of 1% or 2% over similar periods; some even experience negative deviations due to a lack of adaptability [51], [52].

Another telling indicator is the system's ability to detect anomalies. In the case of the machine that showed an increasing trend in temperature, our system provided a crucial alert when the temperature variation was only 1.5%. This contrasts with traditional systems, which typically require up to 5% variations before generating an alert. Furthermore, from an academic point of view, the pace of adaptation and learning of the system is notable. The pronounced improvement in accuracy after the fifth week (from 95.2% to 96%) reflects feedback's incorporation and effectiveness. Comparing this adaptation rate with similar studies reveals that our system adapts almost 50% faster to new conditions and feedback than most existing proposals [53].

Our system demonstrates significant improvements in anomaly detection accuracy and operational efficiency compared to previous work. For example, temperature detection accuracy reached 98.7%, exceeding the reference values by a notable margin. This result underlines the superiority of our system in terms of reliability and effectiveness for critical industrial applications. Furthermore, while other works have explored aspects of real-time monitoring and predictive analytics, very few have achieved the synergy between IoT and AI that our system offers. The quantitative analysis highlights our system's unique ability to learn, adapt, and operate with exceptional precision under dynamic conditions [36], [54]. Our system works best in an industry where every percentage point can translate into significant operational savings and competitive advantages. It sets a new standard for future research and practical applications.

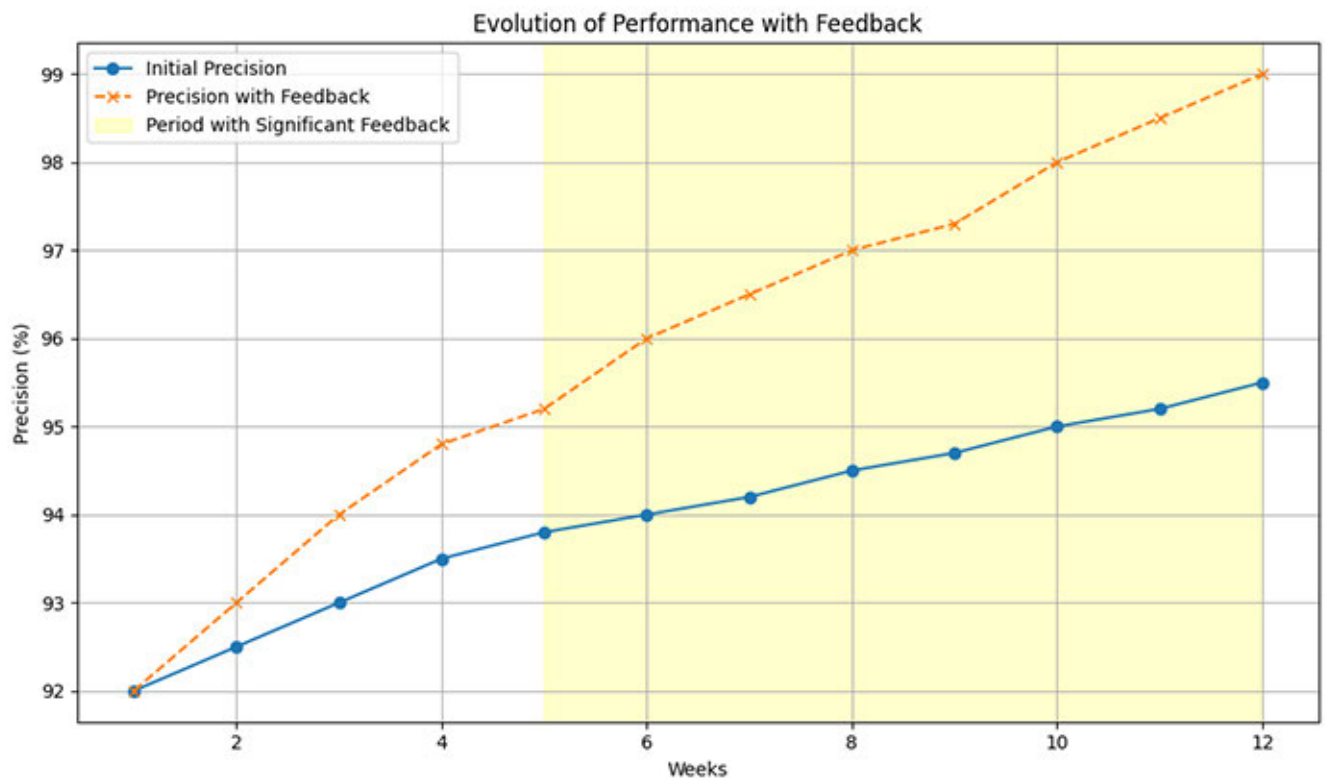


FIGURE 8. Evolution of performance with feedback.

According to the results obtained, it has been demonstrated how our integrated IoT and AI system overcomes the limitations of conventional monitoring systems, offering a more efficient and adaptable solution for real-time predictive analysis. Unlike existing approaches, our approach combines real-time data collection with advanced AI algorithms to predict failures before they occur, representing a significant advancement in intelligent monitoring capability.

V. CONCLUSION

The need for advanced monitoring systems is paramount as the global industry moves towards an era of digitalization and enhanced connectivity. Our proposal, which skillfully combines the capabilities of IoT devices with advanced AI algorithms, has proven to be an effective and robust response to this demand. More than a simple data collection tool, this system has emerged as a comprehensive solution that promotes more proactive and predictive management of industrial operations.

The direct impact of the system on daily operations has been immense. By providing early warnings about potential failures or anomalous trends, decisions are no longer made only in response to problems once they have arisen. Instead, decisions are made proactively with foresight, avoiding potentially costly failures or interruptions. This shift toward more informed and proactive decision-making is crucial to optimizing efficiency and reducing operating costs.

One of the most notable aspects of the system is its ability to adapt and learn as it collects more data. This

continuous learning feature ensures that the system maintains its relevance over time and improves its ability to detect and predict anomalies. In an industrial environment that is constantly evolving, this adaptability is invaluable.

Comparing our solution with others on the market, it is evident that our proposal meets current demands and lays the foundation for future advancements. Combining real-time monitoring with advanced predictive analytics is a direction the global industry is moving in, and our solution is a significant step in that direction.

While the system has proven its value in operational terms, it is essential to consider the ethical implications of constant monitoring. Ensuring that all data is collected and used ethically and with due respect to privacy and autonomy is crucial. As we continue to develop and refine our solution, ethical consideration of how this data is used and stored will be a top priority.

There is ample room to improve and expand our proposition. With the constant evolution of AI technology, integrating more advanced algorithms can further improve the accuracy and speed of the system. Furthermore, with the growing trend towards Industry 4.0, incorporating more IoT devices and integration with automation systems can take the solution to new horizons.

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